INTERNSHIP REPORT

MBBS MENTOR PLATFORM

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ABOUT SCRIPTLANES

Script Lanes is a Pune-based technology company that focuses on delivering AI-driven solutions, web and mobile application development, and SaaS-based services. The company has over 14 years of experience and a combined expertise of more than 50 years. It primarily works with small and medium-sized enterprises (SMEs) to provide customized and scalable technology solution

INTRODUCTION

The MBBS Mentor Platform is an AI-powered web application developed to assist medical students in managing their studies effectively. It provides structured learning support by processing study materials, offering intelligent guidance, and tracking performance. The platform integrates modern technologies to enhance learning efficiency and provide a personalized academic experience.

1. AI-powered notes generation and summarization
2. OCR-based processing of PDF and image study materials
3. Performance tracking with intelligent testing system

OBJECTIVES & SCOPE

Objectives

- Develop a scalable REST API backend
- Enable OCR-based note processing & AI summarization
- Build RAG-based AI assistant for doubt solving
- Implement adaptive testing & analytics system
- Create secure token-based payment system
- Design user-friendly student & admin interfaces

Scope

- Full-stack development (Frontend + Backend)
- AI integration (OCR, embeddings, LLMs)
- Cloud services (AWS S3, Textract, SES)
- Asynchronous processing using queues
- Secure authentication (JWT & OAuth)
- Database management (PostgreSQL, Redis, Pinecone)
- Payment integration (Razorpay)

METHODOLOGY

Requirement analysis & feature identification

Workflow design for AI-based learning system

Note upload → OCR → AI summarization

Token-based usage & payment integration

Test generation & performance evaluation

Query processing using context-based AI

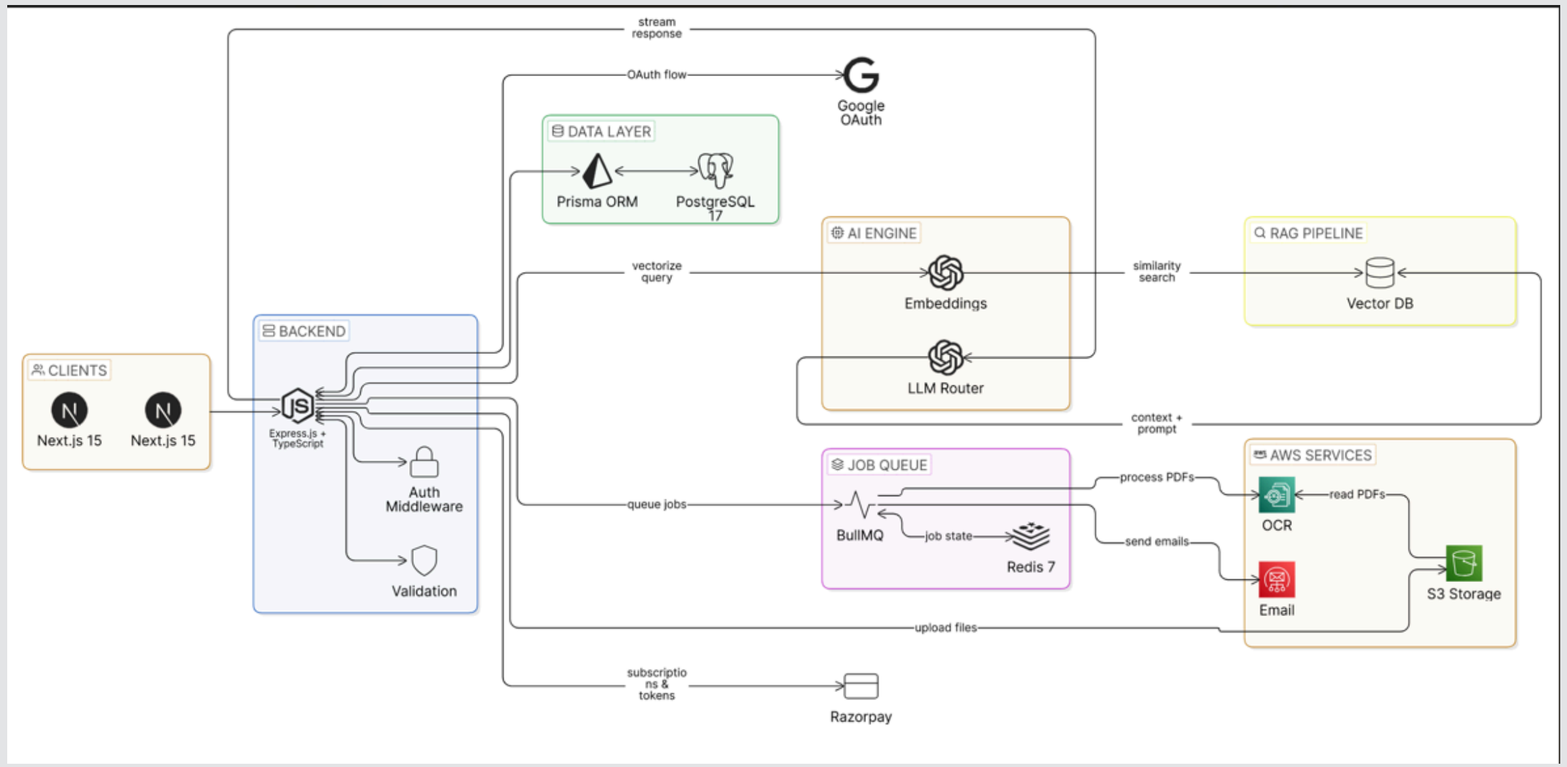
TECHNICAL DETAILS

The MBBS Mentor Platform is developed using a modular full-stack architecture integrating AI and cloud services.

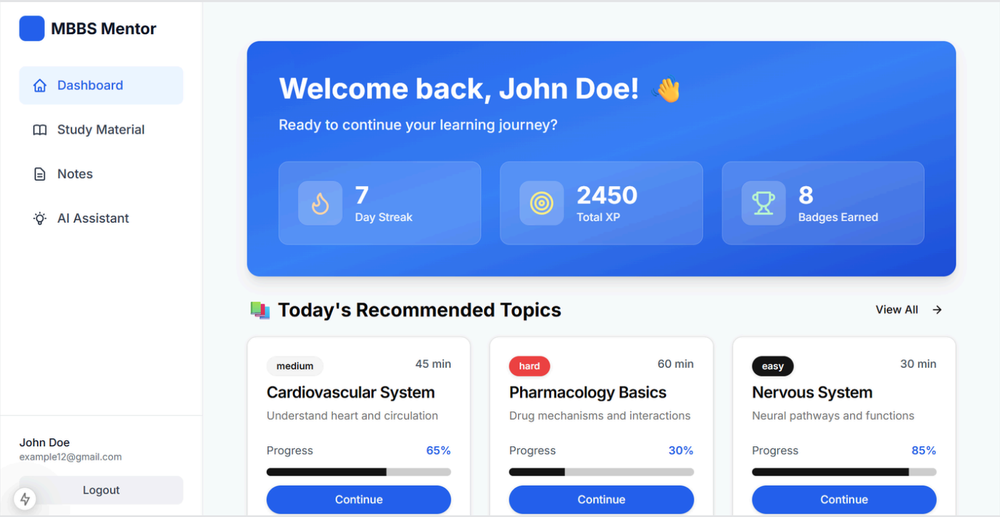
- Frontend: HTML, CSS, JavaScript
- Backend: Flask / Node.js (mention what you used)
- Database: MySQL / PostgreSQL
- OCR Processing: AWS Textract for extracting text from PDFs and images
- AI Integration: Generative AI models for notes generation, summaries, and diagrams
- Payment Integration: Razorpay for subscription management
- APIs & Tools: REST APIs, Postman for testing

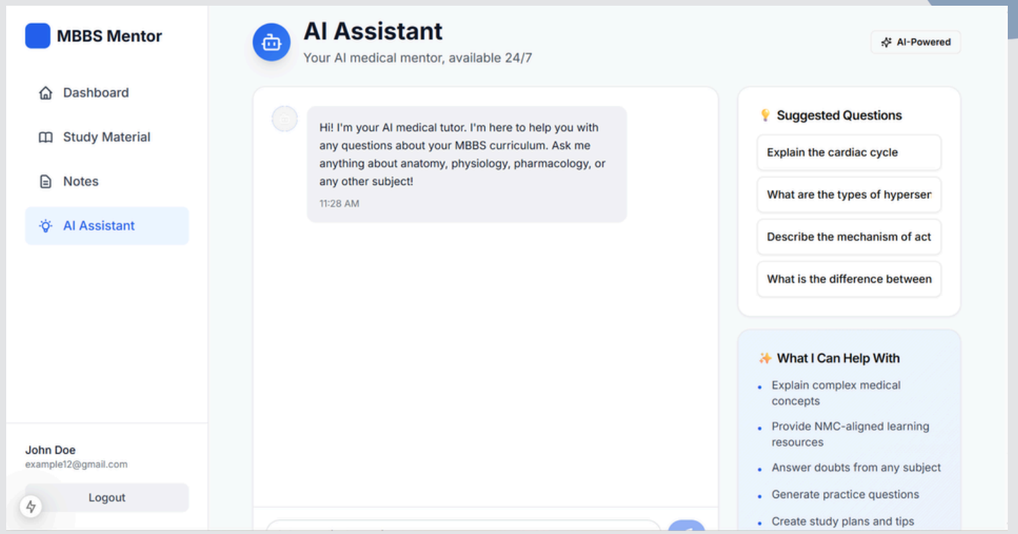
ARCHITECTURE DIAGRAM

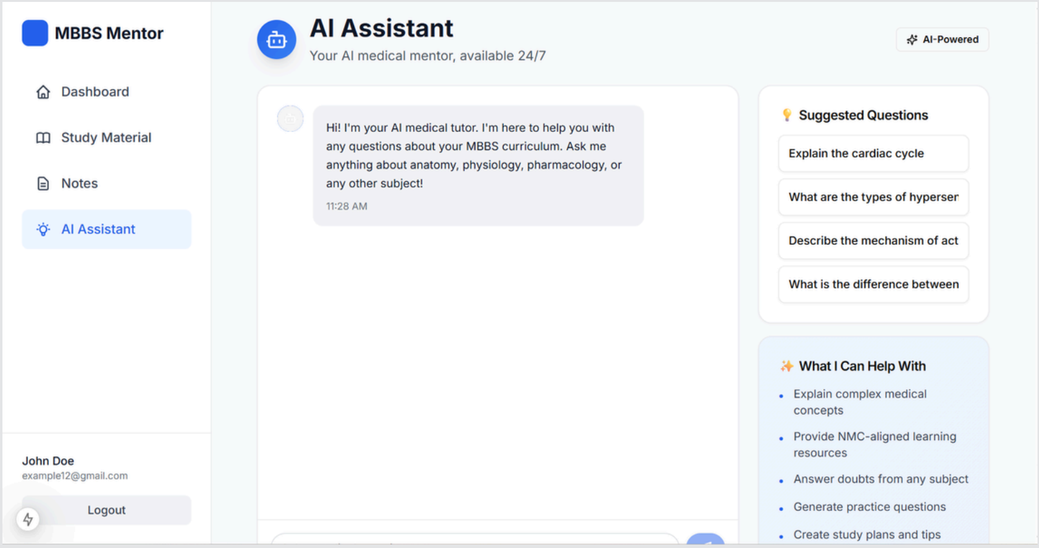
- Student types a question → hits /rag endpoint
- Question converted to a vector → OpenAI embeddings
- Vector searched in Pinecone → finds relevant note chunks
- Matched chunks assembled → context string built
- Question + context sent to LLM → via OpenRouter
- Answer streams back → displayed in chat UI



RESULTS







- Production-ready REST API
- AI note processing pipeline
- RAG chatbot
- Testing + analytics system
- Token economy system
- Dual frontend apps

FUTURE SCOPE

- Multimodal learning (text, images, voice-based AI)
- Virtual patient simulations & clinical decision support
- Advanced personalized learning paths
- Automated revision & weak topic detection
- Integration with global exams (USMLE, PLAB, FMGE)
- Real-time medical database integration (PubMed, WHO)
- Mobile app development & institutional adoption

CONCLUSION

This internship experience helped me gain practical knowledge in full-stack development and AI-based applications. I worked on real-world problems and improved my technical and problem-solving skills. It also strengthened my interest in full-stack and backend development roles. Thank you for your time.

THANK YOU!